
A TALK ON SEQUENCE MODELING

From Words to Worlds

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A journey from bilingual dictionaries to thinking in embeddings.

What this talk is about

Most interesting problems in the world are sequence problems — language, vision, biology, decisions, even thought itself.

This talk traces the arc of how we learned to model sequences: from naïve word-by-word translation, to phrase tables, to RNNs, to Transformers, to Mamba — and asks a final question:



Do our models really need to think in English?

PART ONE

Why Sequence Modeling *is Everything*

Three parts · roughly one hour · one core idea

Not every problem needs a model

Some problems have closed-form answers. No neural network. No training. Just arithmetic.

What's my speed?	$\text{distance} \div \text{time}$	<i>kinematics</i>
Do I have a fever?	$\text{temperature} > 37.5^{\circ}\text{C}$	<i>single threshold</i>
Area of a room?	$\text{length} \times \text{width}$	<i>geometry</i>
Compound interest?	$P(1+r)^n$	<i>deterministic</i>
Final velocity?	$v = u + at$	<i>Newtonian</i>

These problems live inside a textbook. The world doesn't.

But the interesting problems aren't like that

- *Translate this sentence to French.* *language*
 - *What will this protein fold into?* *biology*
 - *What's in this image?* *perception*
 - *What should the agent do next?* *decision*
 - *Will it rain tomorrow?* *forecasting*
-

There is no formula. The input is a sequence, the output is a sequence,
and the mapping must be learned from data.

What is a sequence?

An ordered collection where position matters — swap two elements and the meaning changes.

Text

sequence of tokens

Speech

sequence of audio frames

Video

sequence of images

DNA

sequence of base pairs

Stock prices

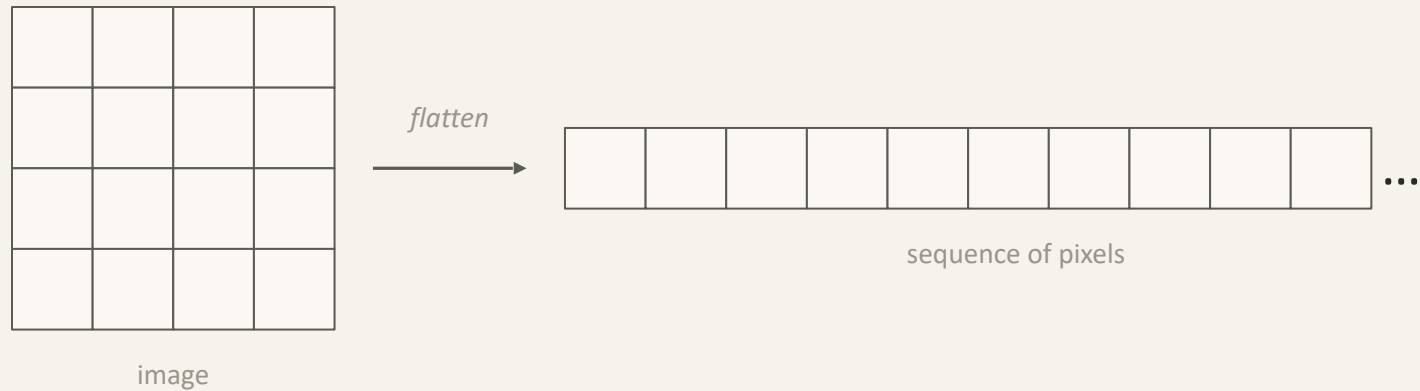
sequence over time

A game of chess

sequence of moves

How vision became a sequence problem

An image is a 2-D grid of pixels — flatten it, and it becomes a sequence of numbers.



Vision Transformers (ViT)

split the image into 16×16 patches and treat each patch as a token.

Audio

spectrograms are images of sound — same trick applies.

Video

a sequence of images — a sequence of sequences.

The four shapes of sequence problems

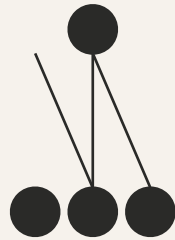


one $\rightarrow$ one

single in, single out

digit classification

the trivial case

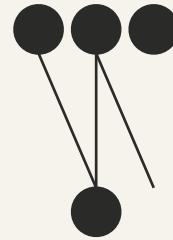


one $\rightarrow$ many

single in, sequence out

image captioning

generation from a fixed input

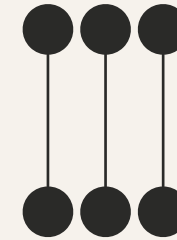


many $\rightarrow$ one

sequence in, single out

sentiment analysis

compress, then decide



many $\rightarrow$ many

sequence in, sequence out

translation, dialogue

the hardest, most general case

Translation sits in the hardest box. The next forty minutes are about how we cracked it.

“

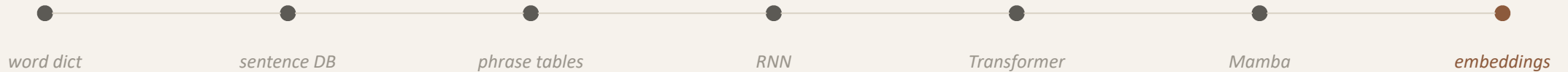
The same machinery that translates English to French also folds proteins, drives cars, and predicts the weather.

So let us take translation as our lens.

PART TWO

Translation as a Lens

the same lessons apply to any sequence problem.



The traveller's instinct

You land in a new country. You don't speak the language. What do you do?

You point at things.

bread → *pain*

dog → *chien*

rain → *pluie*

water → *eau*

You are building a bilingual dictionary — one word at a time. This is humanity's first translation algorithm.

Word-level translation

Translate each word independently using a lookup table, then concatenate. The earliest mechanical translators (1950s) worked this way.

EXAMPLE

cat	→	chat	dog	→	chien	red	→	rouge
the	→	le	run	→	courir	house	→	maison

STRENGTHS

- Simple, transparent, fast — anyone can debug it
- Works reasonably well for nouns and basic vocabulary
- No data, no training required — just a dictionary
- Used in early Cold-War-era machine translation

LIMITATIONS

- “Time flies like an arrow” → word-by-word becomes nonsense
- Idioms break: “kick the bucket” ≠ “frapper le seau”
- Word order is wrong (English SVO vs Japanese SOV)
- No grammar, no agreement, no tense, no context
- One word, many meanings — “bank” is a river or a vault?

→ *We need to look at more than one word at a time.*

From Words to Worlds · The Sequence Modeling Story

Sentence-level translation

Memorise whole sentences and look them up wholesale. This is what tourist phrasebooks have always done.

EXAMPLE

"How are you?" → "Comment allez-vous ?"
"Where is the station?" → "Où est la gare ?"
"I would like a coffee." → "Je voudrais un café."

STRENGTHS

- Grammar and word order preserved naturally
- Sounds fluent and idiomatic on memorised phrases
- Good for tourist phrasebooks and fixed greetings
- Easy to evaluate — exact match works

LIMITATIONS

- Combinatorial explosion — infinitely many sentences exist
- Zero generalisation — change one word and lookup fails
- Cannot say anything you have not seen before
- Storage and search become impractical at scale

→ *We need to translate units smaller than sentences but larger than words.*

Phrase-level translation

The middle ground. Statistical Machine Translation (Koehn et al., ~2003) dominated for over a decade — Google Translate ran on this until 2016.

EXAMPLE

"the red car" → "la voiture rouge"
"is going to" → "va à"
"I would like to" → "je voudrais"

STRENGTHS

- Better generalisation than memorising sentences
- Handles short idioms and common phrasings
- Learnable from parallel corpora using statistics
- Powered every major translation product 2003–2016

LIMITATIONS

- Long-range dependencies still break
- Stitching is brittle — "the X that I saw yesterday" fragments
- Requires huge phrase tables and hand-crafted features
- Cannot handle subject-verb agreement across long distances
- Does not understand meaning — only co-occurrence

→ All three approaches are forms of lookup. None of them read the sentence.

All three share one limitation

Word, sentence, phrase — at heart, all three are forms of lookup.

*They treat translation as lookup,
not as understanding.*

What does “understanding” actually require?

- Read the whole input first
- Build an internal representation
- Generate one piece at a time

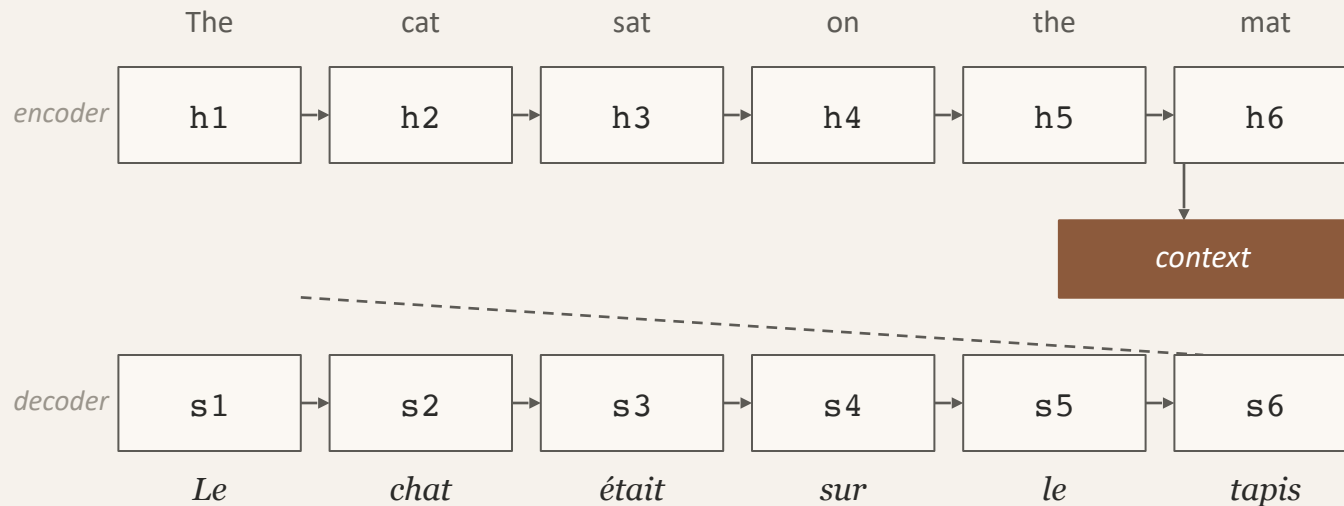
context resolves ambiguity

a vector that captures meaning

conditioned on what was already said

Learning to read: encoder-decoder RNNs

Sutskever et al. (2014). Read the source one token at a time, build a single thought-vector, then unroll it into the target.



STRENGTHS

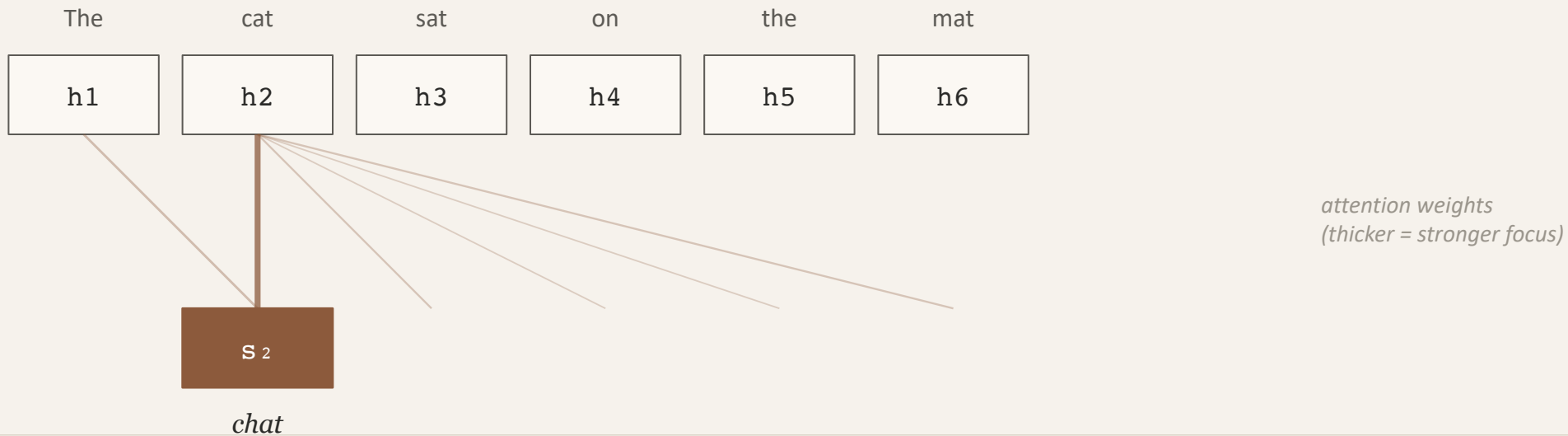
End-to-end learning · variable-length I/O · learns grammar implicitly · LSTM/GRU mitigate vanishing gradients

LIMITATIONS

Sequential — cannot parallelise across time · long sentences lose information through the bottleneck · slow to train

An interlude: letting the decoder peek

Before Transformers, attention was added on top of RNNs to fix the bottleneck.



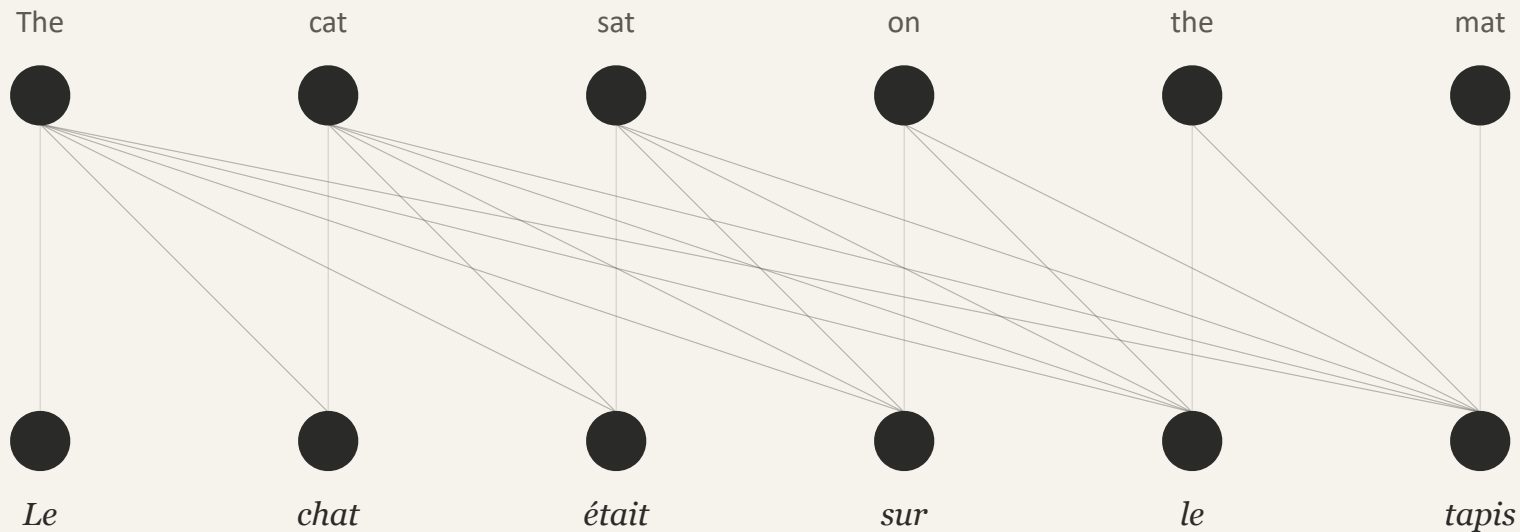
The big idea

When generating each output word, the decoder looks back at all encoder states and decides which ones matter — a soft, learned alignment between source and target.

This idea — that attention itself is the useful part — would prove more important than the recurrence around it.

Attention is all you need

Vaswani et al. (2017). Throw away recurrence entirely. Every token attends to every other token, in parallel.



STRENGTHS

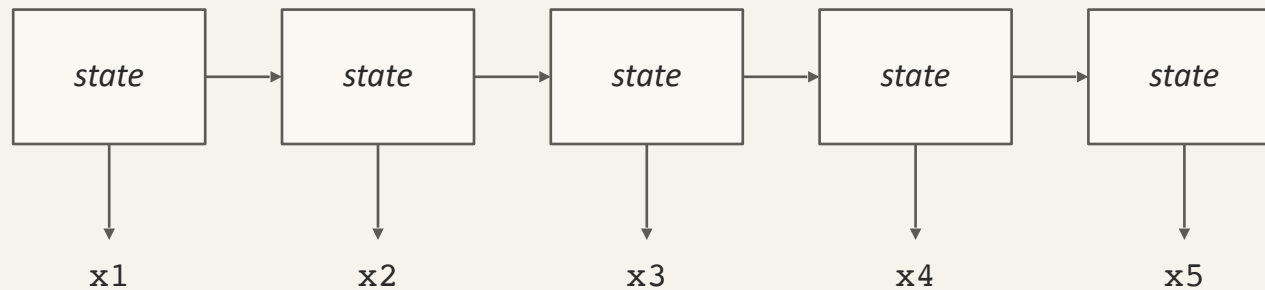
Massively parallel training · long-range dependencies handled directly · one architecture for text, vision, audio, code, biology · the foundation of GPT, BERT, ViT, AlphaFold

LIMITATIONS

Attention is $O(n^2)$ — 1k tokens = 1M comparisons, 100k = 10B · memory scales quadratically · still autoregressive at inference · long-context is expensive

Mamba: bringing back recurrence, cleverly

Gu & Dao (2023). What if we kept the parallel training of Transformers but the linear cost of RNNs?



COMPLEXITY

Transformer: $O(n^2)$

Mamba: $O(n)$

constant memory at inference

A selective state-space decides what to remember and what to forget — at every step.

STRENGTHS

Linear in sequence length · constant inference memory · handles million-token sequences (DNA, audio, long video) · scales gracefully

LIMITATIONS

Newer, less battle-tested · in-context learning still favours attention · hybrid models (Mamba + a few attention layers) often win in practice

The journey so far

ERA	APPROACH	KEY IDEA	BOTTLENECK	WHAT BROKE THROUGH
~1950	Word dictionary	Lookup	No grammar	Reading more than one word
~1990	Sentence database	Memorise	No generalisation	Decomposing into phrases
~2003	Phrase tables (SMT)	Statistical	No long-range structure	Learning representations
~2014	RNN / LSTM	Hidden state	Sequential, forgetful	Direct token-to-token attention
~2017	Transformer	Attention everywhere	$O(n^2)$ cost	Selective recurrence
~2023	Mamba / SSM	Selective state	Still maturing	Skip language altogether?

Each generation kept the gain of the last while removing one bottleneck.

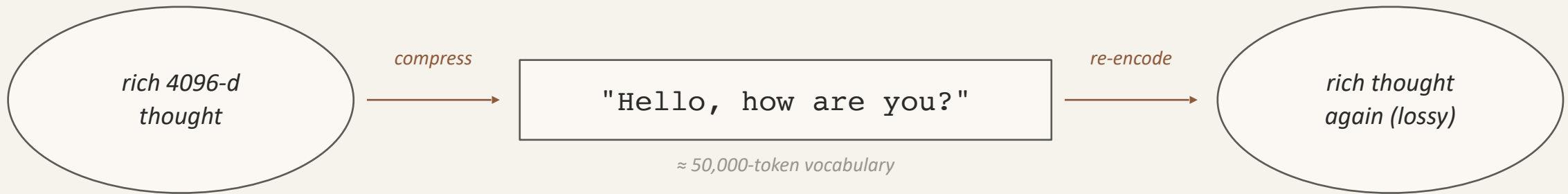
PART THREE

A Final Question

Do agents really need to talk in English?

Language is a lossy compression

When a model thinks, it operates in a high-dimensional vector space. When it speaks to another model, it must compress that thought into a discrete token vocabulary — and the receiver has to decompress.

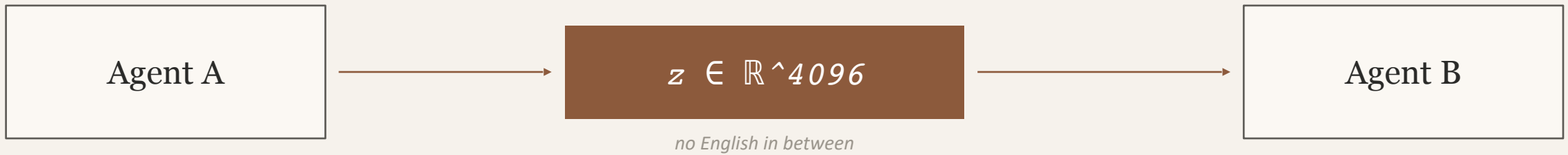


What gets lost

- **Uncertainty** "might", "probably", "somewhat" are coarse buckets
- **Nuance** the model knew more than what fit in the sentence
- **Multimodality** an image-derived thought becomes a paraphrase
- **Continuity** smooth gradients become hard token boundaries

Agents talking in embedding space

What if agents skipped English altogether and exchanged the underlying vectors directly?



CURRENT RESEARCH

V-JEPA (Meta)

predict in latent space, not pixel space — vision without reconstruction

Large Concept Models

operate on “concept” embeddings rather than tokens — sentence-level reasoning

Coconut (Meta)

let models reason in continuous thought-space before emitting words

WHY IT MIGHT MATTER

Faster · richer signal · multimodal-native · closer to how thought actually feels

THE TRADE-OFF

Less interpretable · harder to debug · we lose the ability to read the conversation

Trust the model with more, compress less



The pattern

Every breakthrough kept the gain of the last while loosening one rigid assumption.

The lesson

We learned to trust the model with more of the work — not by hand-crafting structure, but by letting it discover its own.

The open question

How much of language is fundamental to thought, and how much is just the medium we happened to grow up speaking?

THANK YOU

From Words to Worlds.

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Words → phrases → sentences → hidden states → attention → state-spaces → embeddings.

Each step, we trusted the model with more — and asked it to compress less.

Questions?

“The limits of my language mean the limits of my world.” — Wittgenstein

...unless you skip the language altogether.